

LESSON 8-4 • HOMEWORK PACKET • SHEET 1 OF 2

Writing Inequalities

Maryland Standard 6.EE.B.8 • Learning target – I can write an inequality that describes a constraint in a real situation.

REMEMBER

- An **inequality** compares amounts using $<$, $>$, $\leq$, or $\geq$.
- **more than:** $>$ **less than, fewer than:** $<$
- **at least:** $\geq$ **at most, no more than:** $\leq$
- "At least" and "at most" **include** the boundary.

WORKED EXAMPLE

A food pantry needs at least 40 volunteers on Saturday. Write an inequality for v , the number of volunteers.

Step 1 "At least 40" means 40 or more, so 40 is included.

Step 2 Use $\geq$: $v \geq 40$. Test: 40 and 52 work; 38 does not.

Answer $v \geq 40$

PRACTICE*show your work • label units*

1. Write an inequality for each phrase. Use x .

- a. more than 12 _____ b. at most 30 _____ c. fewer than 8 _____
- d. at least 15 _____ e. no more than 6 _____ f. greater than -3 _____
- g. below 0 _____ h. a maximum of 100 _____

2. Write an inequality for each sign. Use the variable given.

Sign	Inequality
Riders must be at least 54 inches tall. (h)	
Speed limit: 35 miles per hour. (s)	
Buy more than 10 items to get a discount. (n)	
Fewer than 120 people may be in this room. (p)	

3. **City Bus.** A city bus can carry no more than 48 riders. Write an inequality for r , the number of riders. Can the bus carry 48 riders? Can it carry 50 riders? Explain.

Inequality _____

4. **Reverse It.** Write a real-world situation for $t < 60$. Name two solutions.

Situation and solutions _____

5. **Same or Different?** Jordan says " x is at most 10" and " x is less than 10" mean the same thing. Is he right? Use the number 10 to explain.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Look for a sign with a limit (speed limit, height rule, room capacity). Write an inequality for it and tell someone at home what it means.

LESSON 8-4 • HOMEWORK PACKET • SHEET 2 OF 2

Representing Inequalities

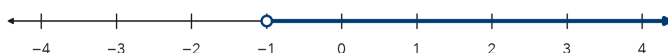
Maryland Standard 6.EE.B.8 • Learning target – I can represent an inequality with a set of solutions and describe them.

REMEMBER

- **Open dot:** boundary **not** included ($<$ or $>$).
- **Closed dot:** boundary **is** included ($\leq$ or $\geq$).
- Arrow: **left** for $<$ and $\leq$, **right** for $>$ and $\geq$.

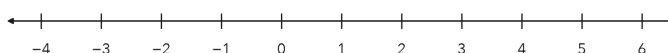
WORKED EXAMPLEGraph $x \leq 3$.**Step 1** $\leq$ includes 3, so draw a closed dot at 3.**Step 2** Shade left: the solutions are 3 and less.**Answer** Closed dot at 3, arrow left**PRACTICE***use the number lines • show your work*

1. Write the inequality shown on the number line. Use x .



Inequality _____

2. Graph $x \geq 2$ on the number line. Then name two solutions.

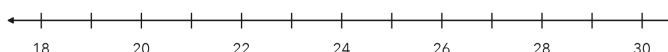


Two solutions _____

3. Each graph is described in words. Write the inequality for each graph. Use x .

a. open dot at 3, arrow right _____**b.** closed dot at -2 , arrow left _____

4. **Water Bottle.** A bottle holds at most 24 ounces of water. Write an inequality for w , the ounces of water. Graph it.



Inequality _____

5. **Find the Error.** Keon graphed $n > 5$ with a closed dot at 5 and an arrow to the right. Explain his mistake. Is 5 a solution?

TEST PRACTICE*choose the best answer*

6. Which inequality is shown on the number line?

**A** $x < 4$ **B** $x \leq 4$ **C** $x > 4$ **D** $x \geq 4$

7. Which number is a solution of $n > -3$?

A -5 **B** -4 **C** -3 **D** 0 **CHECK YOURSELF**☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____